Linear Maps and Models (Lectures 1–4)

Matrix map: $A \in \mathbb{R}^{m\times n}$ maps inputs $x \in \mathbb{R}^n$ to outputs $y = Ax \in \mathbb{R}^m$. With columns a_j and rows r_i^T,

$$y = \sum_{j=1}^{n} x_j a_j, \qquad y_i = r_i^T x, \qquad A_{ij} = e_i^T A e_j$$

Column j is the response to unit input e_j; row i is the sensitivity pattern for output/sensor i.

Products / composition: $ABx = A(Bx)$, so B acts first. Column j of AB is A times column j of B; row i of AB is row i of A times B; and

$$(AB)_{ij} = r_i^T b_j$$

where b_j is column j of B. $(AB)^T = B^T A^T$. Left multiplication mixes rows; right multiplication mixes columns. A permutation matrix has one 1 per row/column, $P^{-1} = P^T$, and reorders rows or columns.

Linear map: $f : \mathbb{R}^n \to \mathbb{R}^m$ is linear iff

$$f(\alpha u + \beta v) = \alpha f(u) + \beta f(v)$$

for all scalars. Necessarily $f(0) = 0$, and every linear map has a unique matrix $A = [f(e_1) \cdots f(e_n)]$. To disprove linearity, one failed test at 0, additivity, or homogeneity suffices.

Affine map: $f(x) = Ax + b$ iff it preserves affine combinations $\alpha + \beta = 1$:

$$f(\alpha u + \beta v) = \alpha f(u) + \beta f(v)$$

Subtract the offset: $g(x) = f(x) - f(0)$ is linear and $b = f(0)$.

Linearization: for differentiable $f : \mathbb{R}^n \to \mathbb{R}^m$,

$$f(x_0 + \delta x) \approx f(x_0) + Df(x_0)\delta x, \qquad (Df)_{ij} = \frac{\partial f_i}{\partial x_j}(x_0)$$

Thus small deviations satisfy $\delta y \approx Df(x_0)\delta x$. For scalar f, $df \approx \nabla f(x)^T dx$, and

$$\nabla(x^T A x) = (A + A^T)x$$

Building engineering models: write one output equation per row and one unknown/input per column. Check dimensions and units. In estimation, rows are sensors and columns are parameter signatures; in control, columns are actuator responses and $\mathrm{range}(A)$ is the set of achievable outputs.

Signal-flow / graph matrices: in a directed linear signal graph, A_{ij} is the sum of products of edge gains over all paths from input j to output i. Cascading the same map gives $A^2, A^3, \ldots$. For adjacency matrix G,

$$(G^k)_{ij} = \#\{\text{walks of length } k \text{ from } j \text{ to } i\}$$

with the orientation determined by the chosen adjacency convention.

Linear Algebra (Lecture 5)

Norms:

$$\|x\|_p = \left(\sum_i |x_i|^p\right)^{1/p}, \qquad \|x\|_\infty = \max_i |x_i|$$

$\|x\|$ means $\|x\|_2$. It is nonnegative with equality iff $x = 0$, $\|\alpha x\| = |\alpha|\|x\|$, and $\|x + y\| \le \|x\| + \|y\|$.

Inner product: $\langle u, v\rangle = u^T v$, or $u^T W v$ for $W = W^T \succ 0$. It is symmetric, linear in each real argument, and positive definite. Its induced norm is $\|u\| = \sqrt{\langle u, u\rangle}$. Key identities:

$$|u^T v| \le \|u\|\|v\|, \qquad u^T v = 0 \Rightarrow \|u + v\|^2 = \|u\|^2 + \|v\|^2$$

$$\cos\theta = \frac{u^T v}{\|u\|\|v\|}$$

Positive inner product means acute ($< 90°$); zero means right; negative means obtuse ($> 90°$).

Hyperplane / halfspace:

$$H = \{x : c^T x = d\}, \qquad H_\pm = \{x : c^T x \gtrless d\}$$

$c \neq 0$ is normal to the hyperplane, whose dimension is $n - 1$. Points closer to a than b form the halfspace

$$\|x - a\| \le \|x - b\| \iff (b - a)^T x \le \frac{\|b\|^2 - \|a\|^2}{2}$$

with perpendicular-bisector boundary through $(a + b)/2$.

Subspace / span: $S \subseteq \mathbb{R}^n$ is a subspace iff $0 \in S$ and $\alpha u + \beta v \in S$ for all $u, v \in S$. The span is

$$\mathrm{span}(v_1, \ldots, v_k) = \left\{\sum_i \alpha_i v_i\right\}$$

Sums and intersections of subspaces are subspaces. $S \cup T$ is a subspace iff $S \subseteq T$ or $T \subseteq S$.

Independence / basis / dimension:

$$\sum_i \alpha_i v_i = 0 \Rightarrow \alpha_i = 0$$

defines independence. A basis of S is an independent set that spans S; every $x \in S$ then has unique coordinates. $\dim S$ is the number of vectors in any basis. In a d-dimensional space, d independent vectors automatically form a basis, as do d spanning vectors.

Coordinates / similarity: for basis matrix $B = [b_1 \cdots b_n]$,

$$x = B\tilde{x}, \qquad \tilde{x} = B^{-1}x$$

If $y = Ax$, its matrix in B-coordinates is

$$\tilde{A} = B^{-1}AB$$

Similar matrices describe the same map in different coordinates and have the same rank, determinant, trace, and eigenvalues.

Invertibility: for square A, the following are equivalent: columns independent; columns span $\mathbb{R}^n$; $\mathrm{null}(A) = \{0\}$; $\mathrm{range}(A) = \mathbb{R}^n$; every $Ax = b$ has one solution; $\det(A) \neq 0$; A^{-1} exists. Equivalently, $\det(A) = 0$ iff A is singular. Useful rules:

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (A^T)^{-1} = (A^{-1})^T$$

Range and Nullspace (Lecture 6)

Definitions:

$$\mathrm{range}(A) = \{Ax : x \in \mathbb{R}^n\} = \mathrm{span}\{\text{columns of } A\}$$
$$\mathrm{null}(A) = \{x : Ax = 0\}$$

$Ax = b$ is consistent iff $b \in \mathrm{range}(A)$. If $z \in \mathrm{null}(A)$, then $A(x + z) = Ax$: nullspace directions are input ambiguity or control freedom.

One-to-one / left inverse: the following are equivalent:

$$\begin{aligned}\mathrm{null}(A) = \{0\} &\iff A \text{ is one-to-one}\\ &\iff \text{columns independent}\\ &\iff \exists L :\ LA = I\end{aligned}$$

For $A \in \mathbb{R}^{m\times n}$ with independent columns,

$$L = (A^T A)^{-1} A^T$$

because $\mathrm{null}(A^T A) = \mathrm{null}(A)$. Conversely, $LA = I$, $Ax = 0 \Rightarrow x = LAx = 0$.

Onto / right inverse: the following are equivalent:

$$\begin{aligned}\mathrm{range}(A) = \mathbb{R}^m &\iff A \text{ is onto}\\ &\iff \text{rows independent}\\ &\iff \exists R :\ AR = I\end{aligned}$$

For independent rows,

$$R = A^T(AA^T)^{-1}$$

Conversely $AR = I$ means $x = Rb$ reaches every b.

Exact solution set: if $Ax = b$ is consistent,

$$x = x_p + z, \qquad z \in \mathrm{null}(A)$$

It is unique iff $\mathrm{null}(A) = \{0\}$, otherwise infinite. A wide matrix has nonzero nullspace, so any consistent system has infinitely many solutions. A tall full-column-rank matrix has at most one solution, but some b are unattainable.

Fundamental orthogonality:

$$\mathrm{range}(A)^\perp = \mathrm{null}(A^T), \qquad \mathrm{range}(A^T)^\perp = \mathrm{null}(A)$$
$$\mathbb{R}^m = \mathrm{range}(A) \oplus \mathrm{null}(A^T),$$
$$\mathbb{R}^n = \mathrm{range}(A^T) \oplus \mathrm{null}(A)$$

Rank (Lecture 7)

Definition / rank-nullity:

$$\mathrm{rank}(A) = \dim \mathrm{range}(A) = \mathrm{rank}(A^T),$$
$$\mathrm{rank}(A) + \dim \mathrm{null}(A) = n$$

Rank is the maximum number of independent columns or rows and the minimum intermediate dimension in a factorization $A = BC$. Full rank means $\mathrm{rank}(A) = \min(m, n)$.

Shape consequences: tall full rank means independent columns, one-to-one, left invertible. Wide full rank means independent rows, onto, right invertible. Square full rank means invertible.

Product bounds: if $B \in \mathbb{R}^{m\times p}$, $C \in \mathbb{R}^{p\times n}$,

$$\mathrm{rank}(B) + \mathrm{rank}(C) - p \le \mathrm{rank}(BC),$$
$$\mathrm{rank}(BC) \le \min\{\mathrm{rank}(B), \mathrm{rank}(C)\}$$

Invertible pre/post multiplication preserves rank.

Block identities:

$$\mathrm{rank}([A\ B]) = \dim(\mathrm{range}(A) + \mathrm{range}(B))$$
$$\mathrm{null}\begin{bmatrix}A\\B\end{bmatrix} = \mathrm{null}(A) \cap \mathrm{null}(B)$$
$$\mathrm{rank}\,\mathrm{diag}(A, B) = \mathrm{rank}(A) + \mathrm{rank}(B)$$

$\mathrm{rank}([A\ B]) = \mathrm{rank}(B)$ iff $\mathrm{range}(A) \subseteq \mathrm{range}(B)$.

Range intersection: if $Z = [Z_A^T\ \ Z_B^T]^T$ spans $\mathrm{null}([A\ -B])$, then $\mathrm{range}(AZ_A) = \mathrm{range}(A) \cap \mathrm{range}(B)$. Retain independent columns of AZ_A, e.g. by QR.

Dimension identities:

$$\dim(S + T) = \dim S + \dim T - \dim(S \cap T)$$
$$(S + T)^\perp = S^\perp \cap T^\perp, \qquad (S \cap T)^\perp = S^\perp + T^\perp$$

$S \oplus T$ means $S + T$ with unique decomposition, equivalently $S \cap T = \{0\}$.

Orthogonality and QR (Lectures 8–9)

Orthonormal columns: $U = [u_1 \cdots u_k]$ has orthonormal columns iff $U^T U = I_k$. Then

$$\|Ux\| = \|x\|, \qquad U^T y = (u_1^T y, \ldots, u_k^T y)$$

If U is square, $UU^T = I$ and $U^{-1} = U^T$. If U is tall, $UU^T \neq I$ in general.

$$\|U^T y\|^2 = \sum_i (u_i^T y)^2 \le \|y\|^2$$

This is Bessel's inequality; equality holds iff $y \in \mathrm{range}(U)$. For square U, equality for every y is Parseval's identity.

2×2 orthogonal maps: $\det U = +1$ means a rotation; $\det U = -1$ means a reflection. Orthogonal products and inverses are orthogonal.

Orthogonal projection of a target: let $S = \mathrm{range}(U)$, $U^T U = I$. For target y, the unique closest point $p \in S$ and perpendicular residual $e \in S^\perp$ are

$$p = P_S y = UU^T y, \qquad e = y - p = (I - P_S)y,$$
$$U^T e = 0, \qquad y = p + e, \qquad p \perp e$$

$P_S = P_S^T = P_S^2$. If A is any full-column-rank basis for S,

$$P_S = A(A^T A)^{-1} A^T = AA^\dagger$$

Thus $P_S y = y \iff y \in S$, and $P_S y = 0 \iff y \in S^\perp$.

$$\mathrm{range}(I - P_S) = \mathrm{null}(P_S) = S^\perp$$

$\|P_S y\| \le \|y\|$, with equality iff $y \in S$. If orthogonal projectors P, Q commute, PQ projects onto $\mathrm{range}(P) \cap \mathrm{range}(Q)$.

Projector recognition / complements: $P^2 = P$ means projection (possibly oblique); adding $P^T = P$ makes it orthogonal. Orthogonal projectors have eigenvalues 0, 1, so

$$\mathrm{rank}(P) = \mathrm{tr}(P), \qquad P_{S^\perp} = I - P_S, \qquad P_S P_{S^\perp} = 0$$

For a unit direction q,

$$P_{\mathrm{span}(q)} = qq^T, \quad P_{q^\perp} = I - qq^T, \quad H = I - 2qq^T$$

H reflects through $q^\perp$: the q-component changes sign and all perpendicular components stay fixed.

Projection dictionary: LS projects output target y onto $\mathrm{range}(A)$; least norm projects 0 onto the affine solution set $\{x : Ax = y\}$; PCA projects centered data onto a top-PC subspace. For full-row-rank C, projection of target x_0 onto $\{x : Cx = d\}$ is

$$x^\star = x_0 - C^T(CC^T)^{-1}(Cx_0 - d)$$

With metric $\|z\|_W^2 = z^T W z$, $W \succ 0$, replace this by

$$x^\star = x_0 - W^{-1}C^T(CW^{-1}C^T)^{-1}(Cx_0 - d)$$

For one hyperplane $a^T x = b$,

$$x^\star = x_0 - \frac{a^T x_0 - b}{\|a\|^2}a, \qquad \mathrm{dist}(x_0, H) = \frac{|a^T x_0 - b|}{\|a\|}$$

Gram–Schmidt: for independent columns a_j,

$$\tilde{q}_j = a_j - \sum_{i<j}(q_i^T a_j)q_i, \qquad q_j = \tilde{q}_j/\|\tilde{q}_j\|$$

This preserves every leading span:

$$\mathrm{span}(q_1, \ldots, q_j) = \mathrm{span}(a_1, \ldots, a_j)$$

Thin QR:

$$A = QR, \qquad Q^T Q = I, \qquad R \text{ upper triangular}$$
$$r_{ij} = q_i^T a_j\ (i < j), \qquad r_{jj} = \|\tilde{q}_j\|$$

For independent columns R is invertible and $\mathrm{range}(Q) = \mathrm{range}(A)$. If a residual is zero, skip that dependent column; the number of generated q_i's is $\mathrm{rank}(A)$.

Full / pivoted QR:

$$A = [Q_1\ Q_2]\begin{bmatrix}R\\0\end{bmatrix}, \qquad \mathrm{range}(Q_2) = \mathrm{range}(A)^\perp$$

Pivoted QR uses $AP = QR$ to move independent/informative columns first. Gram–Schmidt depends on column order. Replacing a_j by cumulative sums preserves leading spans and Q up to signs, but changes R.

QR solves: for full-column-rank $A = QR$,

$$Ax = b \text{ consistent} \Rightarrow Rx = Q^T b, \qquad P_{\mathrm{range}(A)} = QQ^T$$

The same triangular equation gives least squares when b is inconsistent.

Homework 1–3 Patterns

Matrix construction: translate each output formula into one row. Downsample $y_i = x_{2i}$ uses row e_{2i}^T; pair averaging uses $\frac{1}{2}(e_{2i-1} + e_{2i})^T$. For path systems, multiply gains along a path and add gains across parallel paths.

PSD/null trick: if $M = N^T N$,

$$x^T M x = \|Nx\|^2, \qquad \mathrm{null}(M) = \mathrm{null}(N)$$

For a sum $\sum_i N_i^T N_i$, its nullspace is $\bigcap_i \mathrm{null}(N_i)$. This proves invertibility by showing the only vector annihilated by every term is zero.

Nonnegative-independence trick: if every $u_i \neq 0$ and $u_i^T u_j \ge 0$, then

$$\sum_i \alpha_i u_i = 0, \quad \alpha_i \ge 0 \quad \Rightarrow \quad \alpha_i = 0\ \forall i$$

because the squared norm is a sum of nonnegative terms and includes $\alpha_i^2\|u_i\|^2$.

Rational interpolation:

$$f(t) = \frac{a_0 + \cdots + a_m t^m}{1 + b_1 t + \cdots + b_m t^m}$$

Each sample (t_i, y_i) gives the linear row

$$[1 \quad t_i \quad \cdots \quad t_i^m \quad -y_i t_i \quad \cdots \quad -y_i t_i^m]\,\theta = y_i$$

Increase m until consistent; then verify residual, actual interpolation error, and nonzero denominators at all samples.

Wireless feasibility: with desired common SINR S, direct gains $D = \mathrm{diag}(G_{ii})$, interference $K = G - D$, and noise $\sigma\mathbf{1}$,

$$(D - SK)p = S\sigma\mathbf{1}$$

For each S, solve once and check $0 \le p_i \le p_i^{\max}$; this is a feasibility test, not an optimization problem.

Dynamics / coin collection: piecewise-constant forces give linear positions $x(t) = \sum_j a_{tj} f_j$. Collecting selected targets is $Af = x_{\text{target}}$; consistency decides feasibility. To identify one target that must be missed, delete each target row and test the remaining system.

Finite fault search: when each candidate fault determines a model matrix, solve the linear system for every candidate and compare its predicted scalar measurement with the observed one. Finite model choices can be identified even from one scalar when their predictions differ.

Least Squares (Lecture 10)

Problem / residual: for $A \in \mathbb{R}^{m\times n}$, usually $m > n$,

$$x_{\text{ls}} = \arg\min_x \|Ax - y\|^2, \qquad r_{\text{ls}} = Ax_{\text{ls}} - y$$

Geometrically, Ax_{ls} is the closest point in $\mathrm{range}(A)$ to y. Exact fit occurs iff $y \in \mathrm{range}(A)$.

Normal equations: differentiating gives

$$A^T(Ax_{\text{ls}} - y) = 0, \qquad A^T A x_{\text{ls}} = A^T y$$

Thus $r_{\text{ls}} \perp \mathrm{range}(A)$. If A has independent columns,

$$x_{\text{ls}} = (A^T A)^{-1} A^T y = A^\dagger y$$

The minimizer is unique iff $\mathrm{null}(A) = \{0\}$. If A is rank deficient, the fitted vector Ax_{ls} is still unique, but the

coefficient vector need not be.

Projection: for full column rank A,

$$P_A = A(A^TA)^{-1}A^T, \qquad \hat{y} = P_A y, \qquad r = (P_A - I)y$$

$P_A = P_A^T = P_A^2$, range(P_A) = range(A), and null(P_A) = null(A^T). Also

$$y = P_A y + (I - P_A)y, \qquad P_A y \perp (I - P_A)y$$

For arbitrary rank, $P_A = AA^\dagger$ and $x = A^\dagger y$ selects the minimum-norm coefficient vector producing the same closest fit.

Completion of squares: for any x,

$$\|Ax - y\|^2 = \|r_{\text{ls}}\|^2 + \|A(x - x_{\text{ls}})\|^2$$

because $r_{\text{ls}} \perp A(x - x_{\text{ls}})$. This proves optimality and, when A has zero nullspace, uniqueness.

QR solution: if $A = QR$ is thin QR with $Q^TQ = I$ and invertible R,

$$x_{\text{ls}} = R^{-1}Q^Ty, \qquad \hat{y} = QQ^Ty, \qquad r = (QQ^T - I)y$$

Solve $Rx = Q^Ty$; do not explicitly form R^{-1} or A^TA.

Least-squares estimation / BLUE: in $y = Ax + v$, a linear estimate $\hat{x} = By$ is unbiased in the noise-free case iff $BA = I$. Then $x - \hat{x} = -Bv$. For full-column-rank A, $A^\dagger$ is the smallest left inverse in Frobenius norm:

$$BA = I \implies \|B\|_F^2 \geq \|A^\dagger\|_F^2$$

With equal, uncorrelated sensor noise this is the best linear unbiased estimator.

Weighted least squares: for positive definite weight W,

$$\min_x (Ax - y)^TW(Ax - y) \quad \Rightarrow \quad x = (A^TWA)^{-1}A^TWy$$

when A^TWA is invertible. Equivalently apply ordinary LS to $W^{1/2}A, W^{1/2}y$. The weighted fitted-output map is

$$P_W = A(A^TWA)^{-1}A^TW$$

$P_W^2 = P_W$, $A^TW(P_Wy - y) = 0$, and $P_W^TW = WP_W$. It is orthogonal in the W-inner product, though generally $P_W^T \neq P_W$ in the Euclidean one.

Unknown interference: for $y = Ax + Bv + w$, set $P^\perp = I - BB^\dagger$. Joint LS over x, v eliminates v:

$$\hat{x} = (P^\perp A)^\dagger P^\perp y$$

It is identifiable noise-free iff $P^\perp A$ has independent columns; for full-column-rank A, iff range$(A) \cap$ range$(B) = \{0\}$.

Multiobjective LS (Lecture 11)

Two objectives: with

$$J_1(x) = \|Ax - y\|^2, \qquad J_2(x) = \|Fx - g\|^2$$

the weighted objective is ordinary least squares:

$$J_1 + \mu J_2 = \left\| \begin{bmatrix} A \\ \sqrt{\mu}F \end{bmatrix} x - \begin{bmatrix} y \\ \sqrt{\mu}g \end{bmatrix} \right\|^2, \qquad \mu \geq 0$$

If $A^TA + \mu F^TF$ is invertible,

$$x_\mu = (A^TA + \mu F^TF)^{-1}(A^Ty + \mu F^Tg)$$

The stacked matrix has independent columns iff null$(A) \cap$ null$(F) = \{0\}$ for $\mu > 0$.

Tradeoff curve: x is Pareto optimal if no other point improves one objective without worsening the other. On a plot with horizontal J_2 and vertical J_1, $J_1 + \mu J_2 = \alpha$ has slope $-\mu$. $\mu = 0$ prioritizes J_1; $\mu \to \infty$ prioritizes J_2, then J_1 among minimizers of J_2.

Tikhonov / ridge: with $F = I, g = 0$,

$$x_\mu = (A^TA + \mu I)^{-1}A^Ty, \qquad \mu > 0$$

$A^TA + \mu I \succ 0$ for every A, since

$$z^T(A^TA + \mu I)z = \|Az\|^2 + \mu\|z\|^2 > 0 \quad (z \neq 0)$$

Large μ favors small x; small μ favors data fit.

Push-through identity:

$$(A^TA + \mu I)^{-1}A^T = A^T(AA^T + \mu I)^{-1}$$

It follows by first observing $(A^TA + \mu I)A^T = A^T(AA^T + \mu I)$. This lets you invert the smaller matrix.

Least-Norm Solutions (Lecture 12)

Underdetermined equations: let $A \in \mathbb{R}^{m\times n}$, $m < n$, have full row rank. Every y is attainable and all solutions are

$$x = x_p + z, \qquad z \in \text{null}(A)$$

The unique least-norm solution is

$$x_{\text{ln}} = A^T(AA^T)^{-1}y = A^\dagger y$$

Orthogonality proof: $x_{\text{ln}} \in$ range(A^T) = null$(A)^\perp$. For any solution $x = x_{\text{ln}} + z$, $z \in$ null(A), so

$$\|x\|^2 = \|x_{\text{ln}}\|^2 + \|z\|^2 \geq \|x_{\text{ln}}\|^2$$

x_{ln} is therefore the orthogonal projection of the origin onto the affine solution set $\{x : Ax = y\}$. Also $A^\dagger$ is a right inverse and

$$A^\dagger A \text{ projects onto range}(A^T),$$

$$I - A^\dagger A \text{ projects onto null}(A)$$

Lagrange derivation: for $\min \frac{1}{2}\|x\|^2$ subject to $Ax = y$, use

$$\mathcal{L}(x, \lambda) = \tfrac{1}{2}x^Tx + \lambda^T(Ax - y)$$

Stationarity gives $x = -A^T\lambda$; feasibility gives $\lambda = -(AA^T)^{-1}y$.

Equality-constrained least squares:

$$\min_x \tfrac{1}{2}\|Ax - b\|^2 \quad \text{subject to } Cx = d$$

has KKT equations

$$\begin{bmatrix} A^TA & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} A^Tb \\ d \end{bmatrix}$$

The KKT matrix is invertible iff C has independent rows and null$(A) \cap$ null$(C) = \{0\}$. Proof: multiply the first homogeneous KKT equation by x^T and use $Cx = 0$ to obtain $\|Ax\|^2 = 0$.

Explicit constrained correction: if $H = A^TA \succ 0$, let $x_{\text{ls}} = H^{-1}A^Tb$. Then

$$x^\star = x_{\text{ls}} + H^{-1}C^T(CH^{-1}C^T)^{-1}(d - Cx_{\text{ls}})$$

It is the smallest H-metric correction that enforces $Cx = d$.

Homework 4–5 Patterns

Matrix system identification: with samples as columns, $X = [x^{(1)} \cdots x^{(N)}]$, $Y = [y^{(1)} \cdots y^{(N)}]$,

$$\min_A \|AX - Y\|_F^2, \qquad \hat{A} = YX^T(XX^T)^{-1}$$

when X has full row rank. Each row of A is an independent LS fit.

Multiplicative model: for $\hat{L} = \alpha \prod_j z_j^{\beta_j}$, take logs:

$$\log \hat{L} = \log \alpha + \sum_j \beta_j \log z_j$$

Fit $(\log \alpha, \beta_j)$ by LS, then exponentiate. This minimizes squared log error, approximately relative error for a good fit.

Tomography / inverse problems: line integral j is $y_j = \sum_i \ell_{ij}x_i + v_j$, so $A_{ji} = \ell_{ij}$ and $\hat{x} = A^\dagger y$. Always check residual scale and reshape using the given pixel ordering.

Curve smoothing: second-difference matrix

$$(Dg)_i = \frac{g_{i+2} - 2g_{i+1} + g_i}{h^2}$$

penalizes curvature. For

$$\frac{1}{n}\|f - g\|^2 + \frac{\mu}{n-2}\|Dg\|^2$$

the solution satisfies

$$\left(\frac{1}{n}I + \frac{\mu}{n-2}D^TD\right)g = \frac{1}{n}f$$

$\mu = 0 \Rightarrow g = f$; $\mu \to \infty \Rightarrow Dg = 0$, so g is the best affine fit to f.

Interpolation basis: if $B \in \mathbb{R}^{n\times m}$ has independent columns, select m independent rows to form invertible Z, then

$$V = BZ^{-1}$$

has the same range and identity rows at the selected sample indices. Consequently every $y \in$ range(B) is reconstructed directly from those m sampled entries.

Minimum-energy flow/control: encode conservation or endpoint requirements as $Cf = d$. The minimum-energy input is

$$f^\star = C^T(CC^T)^{-1}d$$

for independent constraints. Incidence matrices turn node conservation into $Af = -s$.

Conservation-law correction: to minimally adjust y so $A^Tx = b$, solve $A^Tc = b - A^Ty$:

$$y_{\text{adj}} = y + A(A^TA)^{-1}(b - A^Ty)$$

This is orthogonal projection onto the affine set $\{x : A^Tx = b\}$.

Complex Numbers / Complex Linear Algebra

Arithmetic: $i^2 = -1$, $z = a + ib$, $\bar{z} = a - ib$,

$$|z| = \sqrt{a^2 + b^2} = \sqrt{\bar{z}z}, \qquad z^{-1} = \bar{z}/|z|^2 \quad (z \neq 0)$$

Real-coefficient polynomials, and therefore real matrices, have nonreal roots/eigenvalues in conjugate pairs.

Polar form / Euler:

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta), \quad r = |z|, \quad \theta = \arg z$$

Multiplication multiplies moduli and adds angles:

$$z_1z_2 = r_1r_2e^{i(\theta_1+\theta_2)}, \qquad z^k = r^ke^{ik\theta}$$

$|z| < 1$ gives decay, $|z| > 1$ growth, and $|z| = 1$ constant magnitude with rotation/oscillation.

Complex vectors: $u^* = \bar{u}^T$,

$$\langle u, v\rangle = u^*v, \qquad \|u\|^2 = u^*u$$

Use Hermitian transpose *, not plain transpose, in orthogonality, least squares, and unitary matrices. Complex LS is

$$x_{\text{ls}} = (A^*A)^{-1}A^*y$$

for full column rank.

Real representation: for $A = A_R + iA_I$, $x = x_R + ix_I$,

$$\widetilde{A} = \begin{bmatrix} A_R & -A_I \\ A_I & A_R \end{bmatrix}, \qquad \widetilde{x} = \begin{bmatrix} x_R \\ x_I \end{bmatrix}$$

Then $\widetilde{Ax} = \widetilde{A}\widetilde{x}$, $\|\widetilde{x}\| = \|x\|$, and $\widetilde{A^*} = \widetilde{A}^T$.

Eigenvalues / Diagonalization (Lecture 13)

Definitions:

$$Av = \lambda v, \quad v \neq 0 \iff (A - \lambda I)v = 0,$$

$$\det(\lambda I - A) = 0$$

v is a right eigenvector. A left eigenvector satisfies $w^*A = \lambda w^*$. Eigenvectors may be scaled by any nonzero scalar. The eigenspace is null$(A - \lambda I)$. Its dimension (geometric multiplicity) is at most the root multiplicity of λ in the characteristic polynomial (algebraic multiplicity).

Action / dynamics: along v, A only scales by λ. For $x_{k+1} = Ax_k$, the mode v evolves as λ^kv. If $\lambda = re^{i\theta}$, r controls growth/decay and θ controls rotation/oscillation. The spectral radius is $\rho(A) = \max_i |\lambda_i|$; for diagonalizable A, $A^k \to 0$ iff $\rho(A) < 1$. For real A, a conjugate pair $re^{\pm i\theta}$ produces a real invariant plane spanned by the real and imaginary parts of its eigenvector; on that plane A acts as scaling by r plus rotation.

Diagonalization: $A \in \mathbb{C}^{n\times n}$ is diagonalizable iff it has n independent eigenvectors. With

$$V = [v_1 \ \cdots \ v_n], \qquad \Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$$

$$AV = V\Lambda, \qquad A = V\Lambda V^{-1}$$

Distinct eigenvalues imply independent eigenvectors and diagonalizability; the converse is false. A matrix lacking a full eigenbasis is defective. Rows of V^{-1} are dual left eigenvectors: $w_i^*v_j = \delta_{ij}$. For $x = Vc = \sum_i c_iv_i$, $c = V^{-1}x$, so

$$Ax = \sum_i \lambda_ic_iv_i, \qquad c_i = w_i^*x$$

The coefficients are not generally v_i^*x; that simplification requires an orthonormal eigenbasis. Diagonalizability is equivalent to having enough eigenvectors across all eigenspaces: the geometric multiplicities sum to n.

Useful consequences:

$$A^k = V\Lambda^kV^{-1}, \quad A^{-1} = V\Lambda^{-1}V^{-1} \ (\lambda_i \neq 0)$$

$$f(A) = V \text{ diag}(f(\lambda_i))\, V^{-1}$$

for a polynomial/power-series f. Spectral mapping:

$$Av = \lambda v \quad \Rightarrow \quad f(A)v = f(\lambda)v$$

Also $\text{tr}\, A = \sum_i \lambda_i$, $\det A = \prod_i \lambda_i$, counting algebraic multiplicity.

$$\lambda_i(\alpha A + \beta I) = \alpha\lambda_i(A) + \beta,$$

$$\lambda_i(A^{-1}) = \lambda_i(A)^{-1}, \qquad \lambda(A^T) = \lambda(A)$$

The inverse rule requires A invertible. Similarity changes coordinates, not eigenvalues.

AB and BA have the same nonzero eigenvalues: $ABv = \lambda v, \lambda \neq 0 \Rightarrow Bv \neq 0$ and $BA(Bv) = \lambda Bv$.

Nonnormal caution: for diagonalizable $A = V\Lambda V^{-1}$,

$$\|A^k\| \leq \|V\|\,\|V^{-1}\| \max_i |\lambda_i|^k$$

Nearly dependent eigenvectors make V ill-conditioned, so substantial transient amplification can occur even when all $|\lambda_i| < 1$. Symmetric matrices avoid this because $V = Q$ is orthogonal.

Matrix roots / logarithms: for diagonalizable $A = V\Lambda V^{-1}$, choose scalar values $\mu_i^2 = \lambda_i$ or $e^{\nu_i} = \lambda_i$:

$$A^{1/2} = V \text{ diag}(\mu_i)V^{-1}, \qquad \log A = V \text{ diag}(\nu_i)V^{-1}$$

The logarithm requires A invertible. Branches make these nonunique. A real diagonalizable matrix has a real root/log only when choices can be paired conjugately; each distinct negative real eigenvalue must have even multiplicity.

Symmetric Matrices (Lecture 14)

Spectral theorem: if $A = A^T \in \mathbb{R}^{n\times n}$, all eigenvalues are real and an orthonormal eigenbasis exists:

$$A = Q\Lambda Q^T = \sum_{i=1}^n \lambda_iq_iq_i^T, \qquad Q^TQ = I$$

Eigenvectors for distinct eigenvalues are orthogonal; within a repeated eigenspace they can be chosen orthonormal.

Geometric action: Q^Tx resolves x into eigencoordinates, Λ scales each coordinate, and Q reconstructs. Thus a symmetric map has orthogonal input/output principal directions:

$$Ax = \sum_i \lambda_iq_i(q_i^Tx)$$

Each $q_iq_i^T$ is the one-dimensional orthogonal projector onto span$\{q_i\}$.

Quadratic forms: $f(x) = x^TAx$ depends only on

$$A_s = \tfrac{1}{2}(A + A^T), \qquad x^TAx = x^TA_sx$$

For symmetric A, $\|x\| = 1$, and $\lambda_1 \geq \cdots \geq \lambda_n$,

$$\lambda_n \leq x^TAx \leq \lambda_1$$

More generally,

$$\lambda_{\min}\|x\|^2 \leq x^TAx \leq \lambda_{\max}\|x\|^2$$

Indeed, with $c = Q^Tx$,

$$\frac{x^TAx}{x^Tx} = \frac{\sum_i \lambda_ic_i^2}{\sum_i c_i^2}$$

is a weighted average of eigenvalues; the extrema are attained at corresponding eigenvectors.

PSD / PD:

	x^TAx	eigenvalues
$A \succeq 0$	$\geq 0 \ \forall x$	$\lambda_i \geq 0$
$A \succ 0$	$> 0 \ \forall x \neq 0$	$\lambda_i > 0$

$A \succ 0 \Rightarrow A$ invertible and $A^{-1} \succ 0$. $A \succeq 0$ iff $A = F^TF$ for some F. The minimum number of rows of F is rank(A). For a Gram matrix $G = X^TX$, with x_i the columns of X,

$$G_{ij} = x_i^Tx_j, \qquad z^TGz = \|Xz\|^2,$$

$$\text{null}(G) = \text{null}(X), \qquad \text{rank}(G) = \text{rank}(X)$$

For any PSD A, $x^TAx = 0 \iff Ax = 0$.

Indefinite forms / inertia: split the spectral decomposition into positive and negative eigenvalues:

$$A = Q_+\Lambda_+Q_+^T - Q_-\Lambda_-Q_-^T$$

Then

$$x^TAx = \|\Lambda_+^{1/2}Q_+^Tx\|^2 - \|\Lambda_-^{1/2}Q_-^Tx\|^2$$

The minimum row counts are the numbers $n_+(A)$, $n_-(A)$ of positive and negative eigenvalues.

Matrix inequalities: $A \succeq B$ means $A - B \succeq 0$, or $x^TAx \geq x^TBx$ for all x. Useful rules:

$$A \succeq B \Rightarrow -A \preceq -B,$$

$$A \succeq B \Rightarrow C^TAC \succeq C^TBC,$$

$$A \succeq B \succ 0 \Rightarrow A^{-1} \preceq B^{-1}$$

The order is partial. In general $A \succeq B$ does not imply

$A^2 \succeq B^2$.

Symmetric square root: if $A \succeq 0$,

$$A^{1/2} = Q\,\mathrm{diag}(\sqrt{\lambda_i})Q^T, \qquad A^{1/2}A^{1/2} = A$$

For $A \succ 0$, $A^{-1/2}$ exists.

Ellipsoids (Lecture 15)

Quadratic representation: $c \in \mathbb{R}^n$ is the center; for $S = S^T \succ 0$,

$$\mathcal{E} = \{x : (x-c)^T S(x-c) \le 1\}$$

If $Sq_i = \lambda_i q_i$, principal directions are q_i and semiaxis lengths are $1/\sqrt{\lambda_i}$. Large λ_i means a short, thin direction. Boundary parametrization:

$$x = c + Q\Lambda^{-1/2}u, \qquad \|u\| = 1$$

$S \succ 0$ means all $\lambda_i > 0$, so S^{-1} exists and the set is bounded. Singular $S \succeq 0$ instead gives an unbounded elliptical cylinder along $\mathrm{null}(S)$. The longest-to-shortest semiaxis ratio is $\sqrt{\lambda_{\max}(S)/\lambda_{\min}(S)}$.

Scaled / covariance ellipsoid:

$$(x-c)^T\Sigma^{-1}(x-c) \le \alpha$$

has semiaxis lengths $\sqrt{\alpha\lambda_i(\Sigma)}$. Multiplying α by four doubles all lengths.

Image-of-ball representation:

$$\mathcal{E} = \{c + Bz : \|z\| \le 1\}, \qquad B \text{ invertible}$$

is equivalent to $S = (BB^T)^{-1}$. Given S, all valid maps are

$$B = S^{-1/2}R, \qquad R \in \mathbb{R}^{n\times n}, \quad R^TR = I$$

R is any rotation or reflection of the unit ball, explaining why the quadratic matrix S is unique while B is not. The left singular vectors of B give the ellipsoid axes and $\sigma_i(B)$ give their semiaxis lengths. Also

$$\mathrm{vol}(\mathcal{E}) = \mathrm{vol}(\mathbb{B}_n)|\det B| = \frac{\mathrm{vol}(\mathbb{B}_n)}{\sqrt{\det S}}$$

Under an invertible affine map $y = Mx + d$, the center and quadratic matrix transform as

$$c_y = Mc + d, \qquad S_y = M^{-T}SM^{-1}$$

Sensors: if $y = B^Tx$, then

$$\|y\|^2 \le 1 \iff x^TBB^Tx \le 1$$

This is a bounded ellipsoid iff $BB^T \succ 0$, i.e. the sensor vectors span the state space.

Fitting an ellipsoid: $x_k^TSx_k \approx 1$ is linear in the unique symmetric entries of S. A design row contains x_i^2 on diagonal terms and $2x_ix_j$ on off-diagonal terms. Solve

$$\Phi\theta \approx \mathbf{1}$$

then reconstruct S, verify $S \succ 0$, and assess $(x_k^TSx_k - 1)$ using RMS and maximum absolute error.

Matrix Norm and Gain (Lecture 16)

Maximum gain / induced 2-norm:

$$\|A\| = \max_{x\ne0}\frac{\|Ax\|_2}{\|x\|_2} = \max_{\|x\|_2=1}\|Ax\|_2,$$

$$\|A\| = \sqrt{\lambda_{\max}(A^TA)} = \sigma_{\max}(A)$$

The maximizing input is the top eigenvector of A^TA; its output direction is the corresponding left singular vector.

$$\|A\| = \max_{\|u\|=\|v\|=1}|u^TAv|$$

Consequently every entry, row norm, and column norm is at most $\|A\|$: $|A_{ij}| \le \|A\|$, $\|A^Te_i\| \le \|A\|$, $\|Ae_j\| \le \|A\|$.

Minimum gain:

$$\sigma_{\min}(A) = \min_{\|x\|=1}\|Ax\| = \sqrt{\lambda_{\min}(A^TA)}$$

It is zero iff $\mathrm{null}(A) \ne \{0\}$. For a square invertible matrix it is the smallest singular value and $\|A^{-1}\| = 1/\sigma_{\min}(A)$. Equality in the maximum/minimum gain bounds occurs at the corresponding right singular vectors.

Gain bounds / norm properties: for $x \ne 0$,

$$\sigma_{\min}(A) \le \frac{\|Ax\|}{\|x\|} \le \sigma_{\max}(A) = \|A\|,$$

$$\|AB\| \le \|A\|\,\|B\|, \quad \|A^T\| = \|A\|,$$

$$\|\alpha A\| = |\alpha|\|A\|, \quad \|A+B\| \le \|A\| + \|B\|$$

For square X, Y,

$$\sigma_{\max}(XY) \le \sigma_{\max}(X)\sigma_{\max}(Y),$$

$$\sigma_{\min}(XY) \ge \sigma_{\min}(X)\sigma_{\min}(Y),$$

$$\sigma_{\min}(X+Y) \ge \sigma_{\min}(X) - \sigma_{\max}(Y)$$

Frobenius norm:

$$\|A\|_F = \sqrt{\sum_{ij}A_{ij}^2} = \sqrt{\mathrm{tr}(A^TA)} = \sqrt{\sum_i\sigma_i^2}$$

It is the Euclidean norm of all entries, not the maximum gain.

$$\|A\| \le \|A\|_F \le \sqrt{\mathrm{rank}(A)}\,\|A\|$$

Norm invertibility test: $\|E\| < 1$ implies

$$(I-E)^{-1} = \sum_{k\ge0}E^k, \qquad \|(I-E)^{-1}\| \le (1-\|E\|)^{-1}$$

Thus $A + \Delta A$ is invertible if $\|A^{-1}\Delta A\| < 1$.

SVD (Lecture 17)

Thin SVD: every real matrix has an SVD. If $A \in \mathbb{R}^{m\times n}$ has rank r,

$$A = U_r\Sigma_rV_r^T = \sum_{i=1}^r\sigma_iu_iv_i^T$$

$$U_r \in \mathbb{R}^{m\times r}, \quad V_r \in \mathbb{R}^{n\times r},$$

$$\Sigma_r = \mathrm{diag}(\sigma_1,\dots,\sigma_r), \quad \sigma_1 \ge \cdots \ge \sigma_r > 0$$

Columns of U_r, V_r are orthonormal, and

$$Av_i = \sigma_iu_i, \qquad A^Tu_i = \sigma_iv_i$$

For any input,

$$Ax = \sum_{i=1}^r\sigma_i(v_i^Tx)u_i$$

Thus v_i is an input direction, u_i its output direction, and σ_i its gain. The unit ball maps to an ellipsoid with semiaxes σ_iu_i; right-nullspace directions collapse to zero.

$$\|Ax\|^2 = \sum_{i=1}^r\sigma_i^2(v_i^Tx)^2$$

How to construct / recognize it:

$$A^TAv_i = \sigma_i^2v_i, \qquad u_i = \frac{Av_i}{\sigma_i} \quad (\sigma_i > 0)$$

Thus SVD comes from the symmetric EVD of A^TA, but avoids squaring the matrix numerically in actual computation. Orthogonal changes of basis do not change singular values: $\sigma_i(Q_1AQ_2) = \sigma_i(A)$.

SVD and four spaces:

$\mathrm{range}(A)$	$\mathrm{span}\{u_1,\dots,u_r\}$
$\mathrm{range}(A^T)$	$\mathrm{span}\{v_1,\dots,v_r\}$
$\mathrm{null}(A^T)$	$\mathrm{span}\{\text{remaining left } u_i\}$
$\mathrm{null}(A)$	$\mathrm{span}\{\text{remaining right } v_i\}$

$A^TA = V_r\Sigma_r^2V_r^T$ and $AA^T = U_r\Sigma_r^2U_r^T$: nonzero eigenvalues are σ_i^2.

Full SVD:

$$A = U\Sigma V^T, \qquad U \in \mathbb{R}^{m\times m}, \quad V \in \mathbb{R}^{n\times n}$$

with orthogonal U, V and rectangular diagonal Σ, whose first r entries are positive and the rest zero. It means: rotate by V^T, scale/truncate/zero-pad by Σ, then rotate by U.

$$\mathrm{rank}(A) = \#\{\sigma_i > 0\}, \quad \|A\| = \sigma_1, \quad \|A\|_F^2 = \sum_i\sigma_i^2$$

For square A, $|\det A| = \prod_i\sigma_i$. Tall full-column-rank means $r = n$, every input direction has positive gain, and A is one-to-one. Wide full-row-rank means $r = m$, every output is attainable, but $n - m$ input directions lie in $\mathrm{null}(A)$.

Symmetric case: if $A = A^T$, its singular values are the absolute eigenvalues, sorted by magnitude:

$$\{\sigma_i(A)\} = \{|\lambda_i(A)|\}$$

For $\lambda_i \ne 0$, one choice is $v_i = q_i$, $u_i = \mathrm{sign}(\lambda_i)q_i$.

EVD versus SVD: EVD applies to square A, can require complex vectors, and can fail to diagonalize; it maps each eigenvector back onto itself with signed/complex factor λ_i. SVD always exists, uses orthonormal input/output bases, and has nonnegative gains σ_i. For symmetric $A \succeq 0$, they coincide: $U = V = Q$, $\Sigma = \Lambda$.

Moore–Penrose pseudoinverse:

$$A^\dagger = V_r\Sigma_r^{-1}U_r^T$$

Special cases:

$$\text{full column rank: } A^\dagger = (A^TA)^{-1}A^T,$$

$$\text{full row rank: } A^\dagger = A^T(AA^T)^{-1},$$

$$\text{invertible: } A^\dagger = A^{-1}$$

For every A, y, $A^\dagger y$ is the minimum-norm vector among all least-squares minimizers. All LS minimizers are

$$x = A^\dagger y + z, \qquad z \in \mathrm{null}(A)$$

The four identities are

$$AA^\dagger A = A, \qquad A^\dagger AA^\dagger = A^\dagger,$$

$$(AA^\dagger)^T = AA^\dagger, \qquad (A^\dagger A)^T = A^\dagger A$$

Thus

$$AA^\dagger = P_{\mathrm{range}(A)}, \qquad A^\dagger A = P_{\mathrm{range}(A^T)}$$

and $I - AA^\dagger = P_{\mathrm{null}(A^T)}$, $I - A^\dagger A = P_{\mathrm{null}(A)}$.

Regularization limit / filters:

$$A^\dagger = \lim_{\mu\downarrow0}(A^TA + \mu I)^{-1}A^T$$

in the action on data. In singular coordinates,

$$x_\mu = \sum_{i=1}^r\frac{\sigma_i}{\sigma_i^2+\mu}(u_i^Ty)v_i$$

Ordinary LS uses gain $1/\sigma_i$; Tikhonov multiplies it by $\sigma_i^2/(\sigma_i^2+\mu)$, strongly damping small-σ_i directions.

Sensitivity / condition number: on the nonzero singular spectrum, $\kappa = \sigma_1/\sigma_r$; rank deficiency gives $\kappa = \infty$. For invertible A,

$$\kappa(A) = \|A\|\|A^{-1}\| = \frac{\sigma_{\max}}{\sigma_{\min}} \ge 1$$

If $Ax = y$ and only y is perturbed,

$$\frac{\|\delta x\|}{\|x\|} \le \kappa(A)\frac{\|\delta y\|}{\|y\|}$$

A small $\sigma_{\min}$ means large noise amplification and proximity to singularity. The bound is worst-case and is attained by aligned singular directions; roughly $\log_{10}\kappa$ decimal digits can be lost.

Best low-rank approximation: truncate after $k < r$:

$$A_k = \sum_{i=1}^k\sigma_iu_iv_i^T$$

Eckart–Young:

$$\min_{\mathrm{rank}(B)\le k}\|A-B\| = \sigma_{k+1},$$

$$\min_{\mathrm{rank}(B)\le k}\|A-B\|_F = \sqrt{\sum_{i>k}\sigma_i^2}$$

σ_r is the operator-norm distance to rank $< r$; for invertible square A, $\sigma_{\min}$ is distance to singularity. Singular values below the modeling/noise scale are not reliably distinguishable from zero; retaining only values above that floor gives an effective rank and a simpler model.

Truncated-SVD inverse:

$$x_{\mathrm{TSVD}}^{(k)} = \sum_{i=1}^k\frac{u_i^Ty}{\sigma_i}v_i$$

Small k causes bias/oversmoothing; large k admits noise-amplifying directions. Choose k by the error/validation tradeoff, not just numerical rank.

PCA (Lecture 18)

Center first: with data as columns $X = [x_1 \cdots x_N]$,

$$\bar{x} = \frac{1}{N}\sum_ix_i, \qquad X_c = [x_1 - \bar{x} \ \cdots \ x_N - \bar{x}],$$

$$S = \frac{1}{N}X_cX_c^T$$

$S = S^T \succeq 0$. If data are stored as rows, use $S = X_c^TX_c/N$ and transpose the singular-vector roles. Since centered columns sum to zero, $\mathrm{rank}(X_c) \le \min(n, N-1)$. Without centering, the first component can mostly describe the mean rather than variation.

Variance direction: for unit v,

$$\mathrm{var}(v^Tx_i) = v^TSv$$

This is maximized by the top eigenvector u_1 of S, with value λ_1. Subsequent PCs maximize variance subject to orthogonality. The coordinate $u_j^T(x_i - \bar{x})$ is the score of sample i on component j. Total variance is split among the PCs:

$$\mathrm{tr}(S) = \sum_i\lambda_i = \frac{1}{N}\|X_c\|_F^2$$

SVD connection: if $X_c = U\Sigma V^T$, then

$$S = U\frac{\Sigma\Sigma^T}{N}U^T, \qquad \lambda_i = \frac{\sigma_i^2}{N}$$

For column-stored data the principal directions are left singular vectors u_i.

Projection / reconstruction: with $U_k = [u_1 \cdots u_k]$,

$$z_i = U_k^T(x_i - \bar{x}), \qquad \hat{x}_i = \bar{x} + U_kz_i$$

In matrix form,

$$Z = U_k^TX_c = \Sigma_kV_k^T, \qquad \widehat{X}_c = U_kZ = U_kU_k^TX_c$$

The score coordinates are centered and mutually uncorrelated:

$$\frac{1}{N}ZZ^T = \mathrm{diag}(\lambda_1,\dots,\lambda_k)$$

The optimal k-dimensional affine subspace passes through $\bar{x}$ and uses the top PCs. Its mean-square reconstruction error is

$$E_k = \frac{1}{N}\sum_i\|x_i - \hat{x}_i\|^2 = \sum_{i>k}\lambda_i = \frac{1}{N}\sum_{i>k}\sigma_i^2$$

Fraction of variance captured:

$$\rho_k = \frac{\sum_{i=1}^k\lambda_i}{\sum_i\lambda_i} = \frac{\sum_{i=1}^k\sigma_i^2}{\sum_i\sigma_i^2}$$

Pythagoras gives, sample by sample,

$$\|x_c\|^2 = \|U_kU_k^Tx_c\|^2 + \|(I - U_kU_k^T)x_c\|^2$$

so maximizing captured variance and minimizing perpendicular reconstruction error are the same problem. Equivalently, $\widehat{X}_c = U_k\Sigma_kV_k^T$ is the best rank-k approximation. Reconstruction is exact once $k \ge \mathrm{rank}(X_c)$.

Choosing / interpreting k: use a target ρ_k, a scree-plot elbow or spectral gap, and preferably validation for downstream error. Standardize coordinates first when incompatible units/scales should have equal importance; PCA itself is not scale invariant. PC signs are arbitrary. With repeated eigenvalues, the principal subspace is determined but an individual orthonormal basis inside it is not.

Covariance ellipsoid / whitening: for $S \succ 0$,

$$(x-\bar{x})^TS^{-1}(x-\bar{x}) \le \alpha$$

has axes u_i and lengths $\sqrt{\alpha\lambda_i}$. The whitened coordinate $z = S^{-1/2}(x - \bar{x})$ has identity covariance, and

$$\sqrt{(x-\bar{x})^TS^{-1}(x-\bar{x})} = \|z\|$$

is Mahalanobis distance: distance from the mean measured in standard deviations rather than raw units.

Small singular values: for $Y = [y_1 \cdots y_N] \in \mathbb{R}^{m\times N}$, $N \ge m$,

$$\min_{\|q\|=1}\frac{\|q^TY\|}{\|Y\|_F} = \frac{\sigma_{\min}(Y)}{\|Y\|_F}$$

The minimizing q is the bottom left singular vector; a

small value reveals $q^T y_i \approx 0$. Center for affine relations. Spectral gaps above the noise floor reveal effective dimension/source count.

Projection–EVD–SVD–PCA Connections

One recurring operation: resolve into coordinates, weight selected coordinates, then reconstitute:

orthogonal projection	$P_S x = \sum_i (u_i^T x) u_i$
symmetric EVD	$Ax = \sum_i \lambda_i (q_i^T x) q_i$
SVD	$Ax = \sum_i^k \sigma_i (v_i^T x) u_i$
PCA reconstruction	$\hat{x} = \bar{x} + \sum_{i=1}^{k} u_i u_i^T (x - \bar{x})$

A projector is a symmetric matrix whose EVD has only 0's and 1's. Symmetric EVD permits signed weights λ_i; SVD uses nonnegative gains and different input/output directions. PCA applies symmetric EVD to covariance, equivalently SVD to centered data, then keeps the high-variance projectors.

Which target is closest?

$$\text{LS output target } y : AA^\dagger y$$

$$\text{least-norm target } 0 : A^\dagger y$$

$$\text{PCA centered target } x - \bar{x} : U_k U_k^T (x - \bar{x})$$

Always identify the ambient space first: $AA^\dagger \in \mathbb{R}^{m\times m}$ projects outputs, while $A^\dagger A \in \mathbb{R}^{n\times n}$ projects inputs.

Homework 6–7 Patterns

Ellipsoid fitting check: positivity of the fitted eigenvalues proves the quadratic surface is a bounded ellipsoid. Mean residual near zero can hide cancellation; report RMS $\|r - \mathbf{1}\|/\sqrt{N}$ and $\|r - \mathbf{1}\|_\infty$.

Singular-value proof tools: for the square matrices in HW7, use

$$\sigma_{\max}(A) = \max_{\|x\|=1} \|Ax\|, \qquad \sigma_{\min}(A) = \min_{\|x\|=1} \|Ax\|$$

then apply ordinary/reverse triangle inequalities. A row norm is $\|A^T e_i\| \le \sigma_{\max}(A)$; no analogous lower bound holds for every row.

Deconvolution: smoothing operators have smooth, low-frequency right singular vectors at large σ_i, while high-frequency directions often have tiny σ_i. Direct inversion amplifies noise. TSVD discards those directions; Tikhonov damps them continuously.

Tikhonov estimator as a matrix: for

$$B_\mu = (A^T A + \mu I)^{-1} A^T$$

its singular-direction gains are

$$g_i = \frac{\sigma_i}{\sigma_i^2 + \mu}, \qquad \|B_\mu\| = \max_i g_i$$

The g_i need not remain sorted in the order of σ_i. Also

$$AB_\mu - I = -U \operatorname{diag}\left(\frac{\mu}{\sigma_i^2 + \mu}\right) U^T$$

for square full SVD, so worst reconstruction bias is governed by the smallest singular value.

Probability Basics (Lecture 19)

Events / axioms: sample space Ω, outcome $\omega \in \Omega$, event $A \subseteq \Omega$. Probability satisfies

$$\mathbb{P}(A) \ge 0, \qquad \mathbb{P}(\Omega) = 1, \qquad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

for disjoint A, B. Hence

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Conditioning / Bayes:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

A, B are independent iff $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, equivalently $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. For a disjoint exhaustive partition $\{A_i\}$,

$$\mathbb{P}(B) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$$

Distributions: discrete PMF $p(x) = \mathbb{P}(X = x)$, $\sum_x p(x) = 1$. Continuous PDF $f(x) \ge 0$,

$$\mathbb{P}(a \le X \le b) = \int_a^b f(x)\,dx, \qquad \int f = 1$$

CDF $F(t) = \mathbb{P}(X \le t)$ works in both cases; for a continuous variable, $F'(t) = f(t)$ where differentiable. For a joint density $f_{X,Y}$, marginalize via $f_X(x) = \int f_{X,Y}(x,y)\,dy$, and $f_{X|Y}(x \mid y) = f_{X,Y}(x,y)/f_Y(y)$.

Common distributions:

X	$\mathbb{E}X$	$\operatorname{var} X$
$\operatorname{Bern}(p)$	p	$p(1-p)$
$\operatorname{Bin}(n,p)$	np	$np(1-p)$
$\operatorname{Unif}[a,b]$	$(a+b)/2$	$(b-a)^2/12$
$\operatorname{Exp}(\lambda)$	$1/\lambda$	$1/\lambda^2$
$\mathcal{N}(\mu,\sigma^2)$	μ	σ^2

Also $\operatorname{Poisson}(\lambda)$ has mean and variance λ.

Expectation / moments:

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \quad \text{or} \quad \int g(x)f(x)\,dx$$

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}X + b\mathbb{E}Y + c$$

Linearity never requires independence. The kth moment is $\mathbb{E}X^k$; the kth central moment is $\mathbb{E}[(X - \mathbb{E}X)^k]$. Tower property: $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X \mid Y)]$.

Variance:

$$\operatorname{var} X = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2,$$

$$\operatorname{var}(aX + b) = a^2 \operatorname{var} X$$

$$\operatorname{var}(X + Y) = \operatorname{var} X + \operatorname{var} Y + 2\operatorname{Cov}(X,Y)$$

For independent variables the covariance term is zero. Law of total variance:

$$\operatorname{var}(X) = \mathbb{E}[\operatorname{var}(X \mid Y)] + \operatorname{var}(\mathbb{E}[X \mid Y])$$

Covariance / correlation:

$$\operatorname{Cov}(X,Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X\mu_Y$$

$$\rho_{XY} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X\sigma_Y} \in [-1, 1]$$

Independence implies zero covariance; zero covariance generally does not imply independence.

Random vectors: for $x \in \mathbb{R}^n$,

$$\mu = \mathbb{E}x, \qquad \Sigma = \operatorname{Cov}(x) = \mathbb{E}[(x-\mu)(x-\mu)^T] \succeq 0$$

Diagonal entries are variances; off-diagonal entries are covariances. For every deterministic a,

$$a^T\Sigma a = \operatorname{var}(a^T x) \ge 0$$

which proves $\Sigma \succeq 0$ and links covariance eigenvectors to PCA. For deterministic A, b,

$$\mathbb{E}[Ax + b] = A\mu + b, \qquad \operatorname{Cov}(Ax + b) = A\Sigma A^T$$

For cross-covariance $\Sigma_{xy} = \mathbb{E}[(x - \mu_x)(y - \mu_y)^T]$, $\operatorname{Cov}(Ax, By) = A\Sigma_{xy}B^T$. If independent x, w, then $\operatorname{Cov}(Ax + w) = A\Sigma_x A^T + \Sigma_w$. Useful quadratic expectations:

$$\mathbb{E}\|x\|^2 = \operatorname{tr}\Sigma + \|\mu\|^2, \qquad \mathbb{E}[x^T A x] = \operatorname{tr}(A\Sigma) + \mu^T A \mu$$

for symmetric A.

Sample mean / limit theorems: for independent identically distributed X_i with mean μ, variance σ^2,

$$\bar{X}_N = \frac{1}{N}\sum_i X_i, \qquad \mathbb{E}\bar{X}_N = \mu, \qquad \operatorname{var}(\bar{X}_N) = \frac{\sigma^2}{N}$$

LLN: $\bar{X}_N \to \mu$. CLT: for large N, $\bar{X}_N \approx \mathcal{N}(\mu, \sigma^2/N)$.

Gaussian facts: $X \sim \mathcal{N}(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

About $68\%, 95\%, 99.7\%$ lies within $1, 2, 3$ standard deviations. Standardizing gives $Z = (X - \mu)/\sigma \sim \mathcal{N}(0,1)$. Affine transformations of jointly Gaussian vectors are jointly Gaussian. For jointly Gaussian variables, zero covariance implies independence.

Multivariate Gaussian: for $\Sigma \succ 0$,

$$f(x) = \frac{\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)}{(2\pi)^{n/2}\sqrt{\det\Sigma}}$$

Equal-density contours are covariance ellipsoids. If $z \sim \mathcal{N}(0, I)$, then $x = \mu + \Sigma^{1/2}z \sim \mathcal{N}(\mu, \Sigma)$.

Gaussian Conditioning / Practice Final

MMSE meaning: among all functions $g(y)$,

$$\hat{x}_{\mathrm{MMSE}}(y) = \mathbb{E}[x \mid y] = \arg\min_{g(y)} \mathbb{E}\Big[\|x - g(y)\|^2 \mid y\Big]$$

MAP instead maximizes the posterior density. Posterior mean and mode coincide for a Gaussian.

Joint Gaussian conditioning: if

$$\begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix}\right)$$

then

$$\mathbb{E}[x \mid y] = \mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y)$$

$$\operatorname{Cov}(x \mid y) = \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}$$

The conditional covariance does not depend on the observed value y. Moreover,

$$\mathbb{E}[\|x - \hat{x}_{\mathrm{MMSE}}\|^2 \mid y] = \operatorname{tr}(\operatorname{Cov}(x \mid y))$$

For arbitrary variables with finite second moments, the same mean formula is the best affine estimator; it equals the full MMSE estimator when (x, y) is jointly Gaussian. Its error obeys the orthogonality principle

$$\mathbb{E}[(x - \hat{x})(y - \mu_y)^T] = 0$$

Linear Gaussian model: if

$$x \sim \mathcal{N}(0, \tau^2 I), \quad y = Ax + w, \quad w \sim \mathcal{N}(0, \sigma^2 I), \quad x \perp w$$

then

$$\operatorname{Cov}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \tau^2 I & \tau^2 A^T \\ \tau^2 A & \tau^2 AA^T + \sigma^2 I \end{bmatrix}$$

and the MMSE estimator is

$$\hat{x}_{\mathrm{MMSE}} = \tau^2 A^T(\tau^2 AA^T + \sigma^2 I)^{-1} y$$

Its posterior covariance is

$$\Sigma_{x|y} = \left(\tau^{-2} I + \sigma^{-2} A^T A\right)^{-1}$$

Along right singular vector v_i, the posterior variance is

$$\operatorname{var}(v_i^T x \mid y) = \frac{\tau^2\sigma^2}{\sigma^2 + \tau^2\sigma_i^2}$$

Measurements reduce uncertainty most in high-gain singular directions.

MMSE equals regularized LS:

$$\hat{x}_{\mathrm{reg}} = (A^T A + \lambda I)^{-1} A^T y, \qquad \boxed{\lambda = \sigma^2/\tau^2}$$

Use $B(I + CB)^{-1} = (I + BC)^{-1}B$ or the push-through identity. More generally, Gaussian $x \sim \mathcal{N}(\mu_x, \Sigma_x)$ and $w \sim \mathcal{N}(0, \Sigma_w)$ give both MAP and MMSE as the minimizer of

$$(y - Ax)^T\Sigma_w^{-1}(y - Ax) + (x - \mu_x)^T\Sigma_x^{-1}(x - \mu_x)$$

MMSE shrinkage in SVD: for $A = U\Sigma V^T$,

$$\hat{x}_{\mathrm{MMSE}} = \sum_i \frac{\tau^2\sigma_i}{\tau^2\sigma_i^2 + \sigma^2}(u_i^T y)v_i$$

Relative to x_{ls}, component i is multiplied by

$$\gamma_i = \frac{\tau^2\sigma_i^2}{\tau^2\sigma_i^2 + \sigma^2} \in (0, 1)$$

Large gain/low noise/weak prior shrinkage gives $\gamma_i \approx 1$; small gain/high noise/strong small-x prior gives $\gamma_i \approx 0$.

Scalar posterior MSE: for $y = sx + w$ with $s > 0$,

$$\mathbb{E}[(x - \hat{x}_{\mathrm{MMSE}})^2 \mid y] = \frac{\tau^2\sigma^2}{\sigma^2 + s^2\tau^2}$$

It tends to 0 as $\sigma \to 0$ and to τ^2 as $\sigma \to \infty$.

Practice Final Problem Patterns

Warehouse stacking: for two warehouses,

$$A_{\mathrm{ship}} = \begin{bmatrix} I & 0 \\ I & 0 \\ 0 & I \\ 0 & I \end{bmatrix}, \quad y = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \quad A_{\mathrm{pen}} = [I \ \ -I]$$

Use $\hat{A} = [A_{\mathrm{ship}}; \sqrt{\mu}A_{\mathrm{pen}}]$, $\hat{y} = [y; 0]$. Uniqueness holds because A_{ship} has full column rank. At $\mu = 0$, each warehouse is its pair's midpoint; as $\mu \to \infty$, both approach the centroid of all cities. In the practice-final diagonalization, varying μ traces the line segment between these endpoint solutions.

Separate regressions: fitting nonorthogonal feature columns one at a time generally double-counts shared directions and gives a worse residual than joint LS. With orthonormal feature columns U, coefficients decouple:

$$\min_\beta \left\| y - \sum_j \beta_j u_j \right\|^2 \quad \Rightarrow \quad \beta_j = u_j^T y$$

The fitted vector matches joint LS, although coordinates differ from those in the original feature basis.

Rank-deficient regression: zero singular values make $X^T X$ singular. Use $X^\dagger y$; it gives the minimum-norm coefficient among all LS minimizers, while $XX^\dagger y$ is the unique fitted response.

Graph Laplacian: for an undirected weighted graph,

$$L = D - W, \qquad x^T L x = \frac{1}{2}\sum_{i,j} w_{ij}(x_i - x_j)^2 \ge 0$$

Thus $L \succeq 0$, $L\mathbf{1} = 0$, and the multiplicity of eigenvalue zero equals the number of connected components. The graph is connected iff $\operatorname{null}(L) = \operatorname{span}\{\mathbf{1}\}$, equivalently $\lambda_2(L) > 0$.

Decomposition and Model Dictionaries

Which decomposition? QR produces an orthonormal basis for $\operatorname{range}(A)$ plus triangular coordinates; use it for stable solves and LS. **EVD** exposes invariant modes of a square map and simplifies $A^k, f(A)$, but a full eigenbasis may not exist. **Symmetric EVD** always has an orthonormal basis; use it for quadratic forms, PSD tests, covariance, and ellipsoid axes. **SVD** works for every rectangular matrix and exposes gains, rank, four spaces, pseudoinverse, conditioning, and low-rank structure. **PCA** is not a separate factorization: center the data, then use the symmetric EVD of covariance or SVD of the centered matrix.

Singular-direction filter dictionary: let $s_i > 0$ be a singular value and $d_i = u_i^T y$. The coefficient on v_i is

method	v_i coefficient
pseudoinverse	d_i/s_i
TSVD at k	d_i/s_i $(i \le k)$, 0 $(i > k)$
Tikhonov	$s_i d_i/(s_i^2 + \mu)$
Gaussian MMSE	$\tau^2 s_i d_i/(\tau^2 s_i^2 + \sigma_w^2)$

Relative to direct inversion, the last two shrink by

$$\frac{s_i^2}{s_i^2 + \mu}, \qquad \frac{\tau^2 s_i^2}{\tau^2 s_i^2 + \sigma_w^2}$$

Small-s_i modes are precisely where inversion amplifies noise and regularization suppresses it.

Ellipsoid dictionary:

$$\mathcal{E} = \{c + Bz : \|z\| \le 1\} = \{x : (x - c)^T S(x - c) \le 1\},$$

$$S = (BB^T)^{-1}$$

Axes are the left singular vectors of B, with lengths $\sigma_i(B)$. For the covariance ellipsoid $(x - c)^T\Sigma^{-1}(x - c) \le \alpha$, one may take

$$B = \sqrt{\alpha}\,\Sigma^{1/2}, \qquad S = \alpha^{-1}\Sigma^{-1}$$

Thus PCA eigenvectors give the axes, covariance eigenvalues give squared standard deviations, and Mahalanobis distance measures radius in these standardized coordinates.

Four-space projector dictionary: for thin $A = U_r\Sigma_r V_r^T$,

$\operatorname{range}(A)$	$AA^\dagger = U_r U_r^T$
$\operatorname{null}(A^T)$	$I - AA^\dagger = I - U_r U_r^T$
$\operatorname{range}(A^T)$	$A^\dagger A = V_r V_r^T$
$\operatorname{null}(A)$	$I - A^\dagger A = I - V_r V_r^T$

Hence

$$\min_x \|Ax - y\| = \|(I - AA^\dagger)y\|, \qquad \hat{y} = AA^\dagger y$$

Also

$$(A^\dagger)^T = (A^T)^\dagger, \qquad \|A^\dagger\| = \frac{1}{\sigma_r}$$

where σ_r is the smallest positive singular value.

Closest-point proof: verify $p \in S$ and $e = y - p \in S^\perp$. For every $z \in S$, Pythagoras gives

$$\|y - z\|^2 = \|e + (p - z)\|^2 = \|e\|^2 + \|p - z\|^2 \ge \|e\|^2$$

so p is uniquely closest. For an affine set, $P_{x_0+S}(y) = x_0 + P_S(y - x_0)$.

Singular-value perturbation:

$$|\sigma_i(A + E) - \sigma_i(A)| \le \|E\|$$

Normal-equation conditioning:

$$\lambda_i(A^T A) = \sigma_i(A)^2, \qquad \kappa(A^T A) = \kappa(A)^2$$

for full-column-rank A. Normal equations square conditioning and can lose about twice as many digits; prefer QR or SVD numerically.